Mining Media Data I
Winter Semester 26/27
Lecture-3
Affinity Mining II

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Agenda

- Recap: frequent itemsets and FP-Growth
- Association rules
- Beyond confidence: lift
- Visualizing rules
- High utility itemset mining
- Summary

Recap (1): Frequent Itemsets in the Game Shop

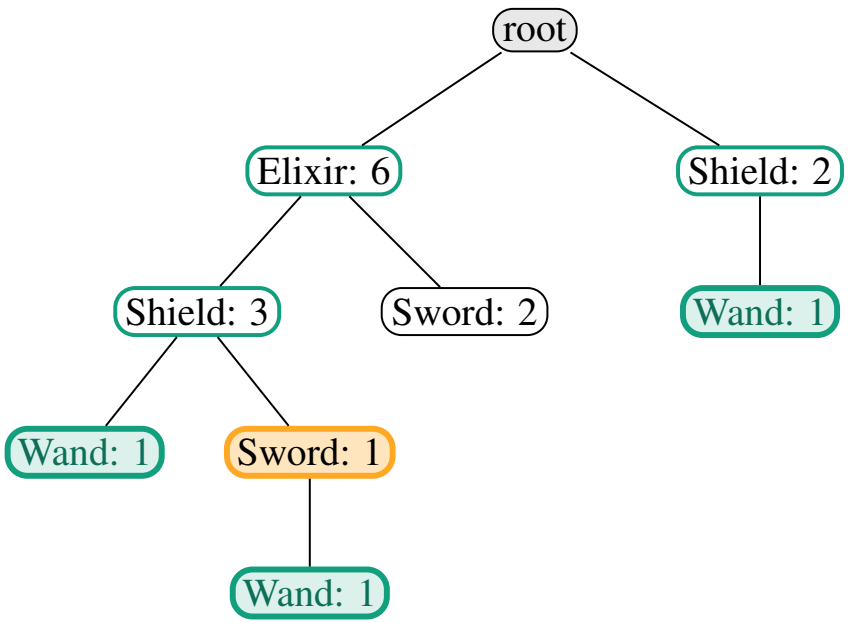
Transaction	**Bought items**
T_1	Elixir, Shield
T_2	Gem, Shield
T_3	Elixir, Sword
T_4	Elixir, Shield, Wand
T_5	Elixir, Sword
T_6	Shield, Wand
T_7	Elixir, Shield, Sword, Wand
T_8	Elixir, Giant Wand

Frequent itemset ($S_{min} = 0.25$)	supp
{Elixir}	0.750
{Shield}	0.625
{Sword}	0.375
{Wand}	0.375
{Elixir, Shield}	0.375
{Elixir, Sword}	0.375
{Shield, Wand}	0.375
{Elixir, Wand}	0.250
{Elixir, Shield, Wand}	0.250

- supp(B): fraction of transactions containing B
- Apriori: subsets of frequent itemsets are frequent

- **Today**: from itemsets to **rules**, and from frequency to **utility**

Recap (2): Open Question on FP-Growth



- Example: frequent itemsets with **Wand**
- Paths to the Wand nodes:
  - Elixir, Shield (1)
  - Elixir, Shield, Sword (1)
  - Shield (1)
- Counts: Shield 3, Elixir 2, Sword 1 $\rightarrow 1/8 < 0.25$

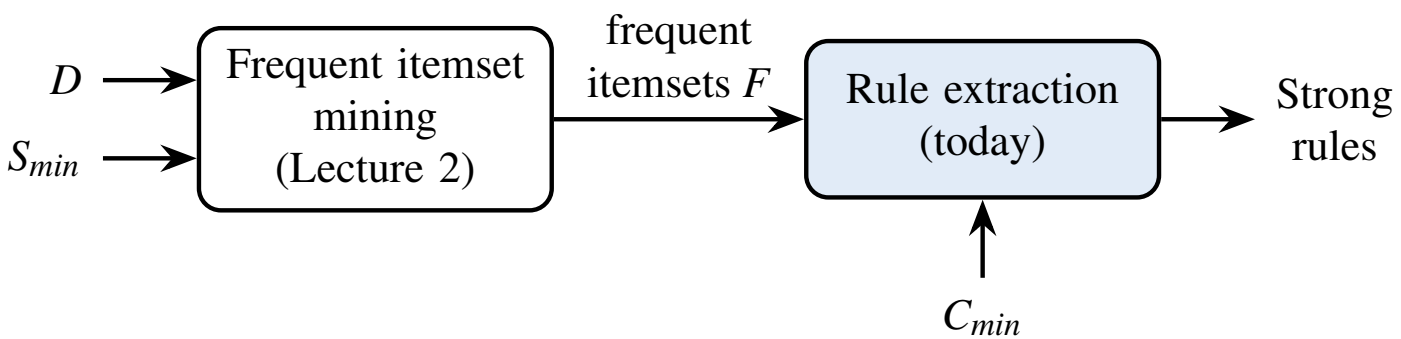
$\Rightarrow$ {Shield, Wand}: 3/8
{Elixir, Wand}: 2/8
{Elixir, Shield, Wand}: 2/8

- Same result as Apriori, **no candidates**

Association Rules: Motivation

- **Frequent itemsets**: *what* is bought together
  - E.g. {Shield, Wand} with supp = 0.375
- The shop needs **if-then** statements
  - A player has a Wand in the basket: offer a Shield?
- Wanted: rules $X \rightarrow Y$ that are
  - **frequent**: enough evidence (S_{min})
  - **reliable**: Y often follows X (C_{min})

$\Rightarrow$ **Strong rules**



Association Rules

- **Association rule** $X \to Y$ with $X, Y \subseteq A$ and $X \cap Y = \emptyset$
  - Read: *who buys X also buys Y*
  - Asymmetric: $X \to Y$ is not the same as $Y \to X$
- **Confidence**: empirical conditional probability $p(Y \mid X)$

$$\text{conf}(X \to Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)}$$

- **Strong rule**: $\text{supp}(X \cup Y) \geq S_{min}$ and $\text{conf}(X \to Y) \geq C_{min}$
- **Example**:
  - $\text{conf}(\text{Wand} \to \text{Shield}) = 0.375/0.375 = 1.0$: every Wand buyer also bought a Shield
  - $\text{conf}(\text{Shield} \to \text{Wand}) = 0.375/0.625 = 0.6$

Association Rules: Direction Matters

	Shield	**Wand**
T_1	✓	
T_2	✓	
T_3		
T_4	✓	✓
T_5		
T_6	✓	✓
T_7	✓	✓
T_8		
$\sum$	5	3

both items

- **Wand → Shield**
  - 3 of 3 Wand buyers also bought a Shield: 100 %
- **Shield → Wand**
  - 3 of 5 Shield buyers also bought a Wand: 60 %
- Same itemset {Shield, Wand}, same support 0.375

$\Rightarrow$ Rules are **asymmetric**:
Wand → Shield $\neq$ Shield → Wand

Rules from One Frequent Itemset

- Frequent itemset $f = \{\text{Elixir}, \text{Shield}, \text{Wand}\}$
- Every nonempty proper subset $X \subset f$ gives a rule $X \to f \setminus X$
- $2^{|f|} - 2 = 6$ rules
- Same **numerator** for all rules: $\text{supp}(f) = 0.25$
  - Only $\text{supp}(X)$ changes
- Keep rules with $\text{conf} \geq C_{min} = 0.5$

Rule	conf
Elixir, Wand → Shield	$0.25/0.25 = 1.00$
Shield, Wand → Elixir	$0.25/0.375 = 0.67$
Elixir, Shield → Wand	$0.25/0.375 = 0.67$
Wand → Elixir, Shield	$0.25/0.375 = 0.67$
Shield → Elixir, Wand	$0.25/0.625 = 0.40$
Elixir → Shield, Wand	$0.25/0.75 = 0.33$

strong rejected

Extracting All Strong Rules

Procedure `ExtractRules`(F, C_{min})**:**

- $R \leftarrow \emptyset$
- **foreach** $f \in F$ *with* $|f| \geq 2$ **do**
  - **foreach** *nonempty* $X \subset f$, $X \neq f$ **do**
    - **if** $\text{supp}(f)/\text{supp}(X) \geq C_{min}$ **then**
      - $R \leftarrow R \cup \{X \rightarrow f \setminus X\}$
- **return** R

- No database scan: the supports of all $X \subset f$ are known

Is Confidence Enough?

- $\text{conf}(\text{Shield} \to \text{Elixir}) = 0.6$: sounds like a good rule
- But: 75 % of *all* transactions contain Elixir anyway

$\Rightarrow$ Shield buyers buy Elixir **less often** than average

Lift

$$\text{lift}(X \to Y) = \frac{\text{conf}(X \to Y)}{\text{supp}(Y)} = \frac{\text{supp}(X \cup Y)}{\text{supp}(X)\,\text{supp}(Y)}$$

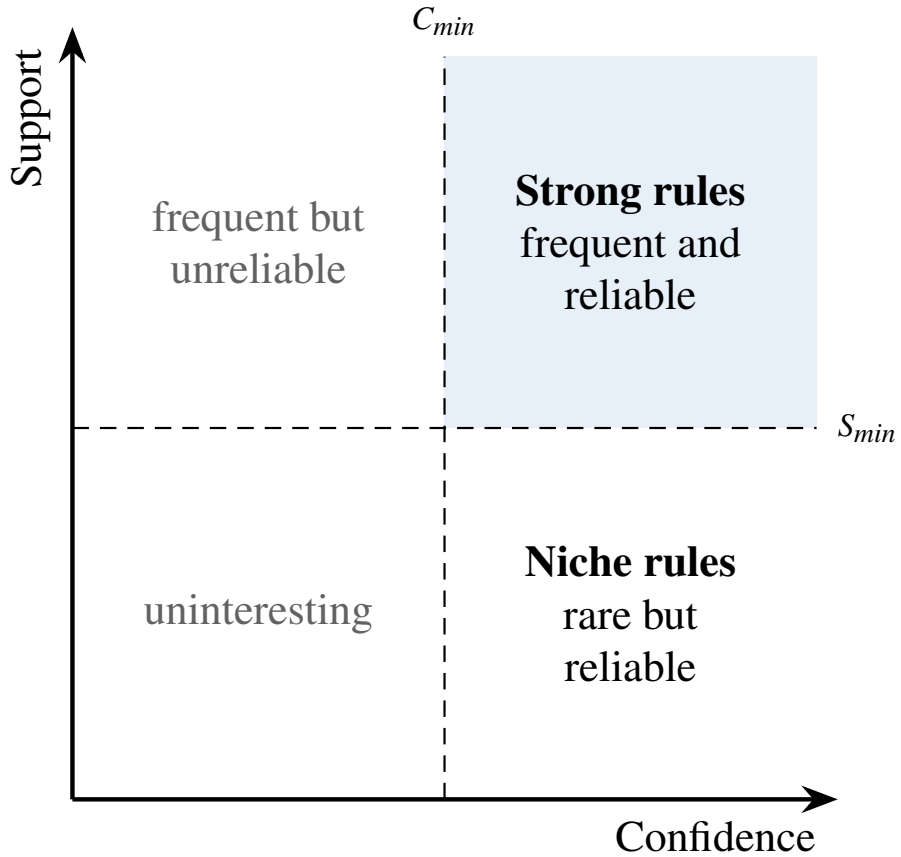
- $\text{lift} > 1$: positively correlated, X makes Y more likely
- $\text{lift} = 1$: independent
- $\text{lift} < 1$: negatively correlated, X makes Y less likely
- Example: $\text{lift}(\text{Shield} \to \text{Elixir}) = 0.6/0.75 = 0.8$

Lift in the Game Shop

Rule	conf	lift
Wand → Shield	1.00	1.60
Shield → Wand	0.60	1.60
Sword → Elixir	1.00	1.33
Elixir → Sword	0.50	1.33
Wand → Elixir	0.67	0.89
Elixir → Wand	0.33	0.89
Shield → Elixir	0.60	0.80
Elixir → Shield	0.50	0.80

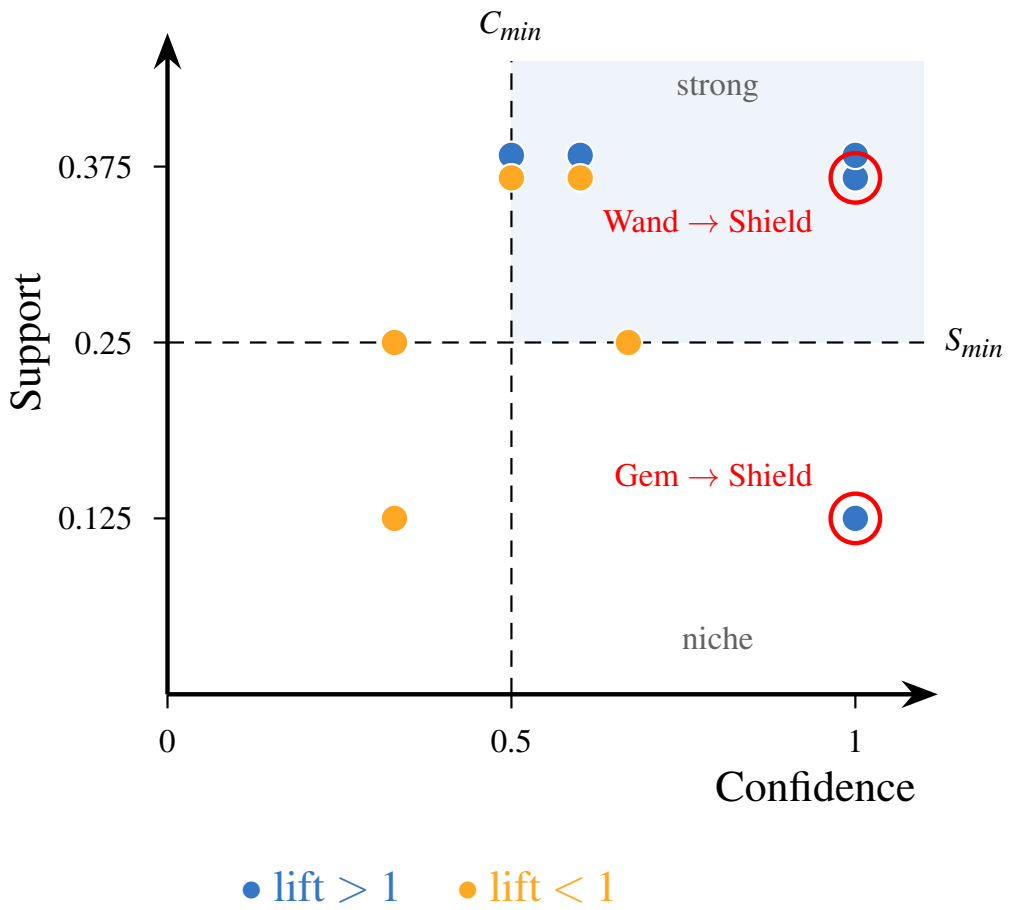
- All rules from frequent 2-itemsets
- Lift is symmetric: $\text{lift}(X \to Y) = \text{lift}(Y \to X)$
- Shield and Wand belong together
- Shield → Elixir: high confidence, but lift < 1
- Use lift to **filter** rules with high confidence

Visualizing Rules: Support–Confidence Plot



- **Strong rules**, e.g. bread $\rightarrow$ butter
- **Niche rules**, e.g. gin, tonic $\rightarrow$ limes
  - Interesting for targeted marketing

Visualizing Rules: Game Shop



- One point per rule
- Color: lift
- **Wand → Shield**
  - supp = 0.375: 3 of 8 baskets
  - conf = 1.0: every Wand buyer took a Shield
  - lift = 1.6: Shield 1.6× as likely as on average

  ⇒ strong rule, e.g. bundle
- **Gem → Shield**
  - Same conf, lift, but supp = 0.125: 1 basket

  ⇒ niche rule, weak evidence

High Utility Itemset Mining: Motivation

- **Quality over quantity**: frequency alone is not always what matters
- Example: marketing wants the itemsets with the **highest revenue**
- So far, a transaction only says *which* items were bought
- Now additionally:
  - **Quantity**: how many of each item
  - **Price** (or profit, rank, ...): how valuable an item is
- In the game shop: Gem and Giant Wand are rare, but expensive

High Utility Itemset Mining: Data

	Items (quantity)	$\mathrm{tu}(T_i)$
T_1	Elixir (1), Shield (1)	2
T_2	Gem (2), Shield (1)	11
T_3	Elixir (2), Sword (2)	4
T_4	Elixir (1), Shield (1), Wand (1)	6
T_5	Elixir (2), Sword (3)	5
T_6	Shield (1), Wand (1)	5
T_7	Elixir (1), Shield (3), Sword (1), Wand (2)	13
T_8	Elixir (5), Giant Wand (3)	20

Item	**Price**
Elixir	1
Gem	5
Giant Wand	5
Shield	1
Sword	1
Wand	4

- Same items as before, now with quantities and prices
- $\mathrm{tu}(T_i)$: **transaction utility**, the total revenue of T_i

Utility (1): Within One Transaction

	Items (quantity)
T_1	Elixir (1), Shield (1)
T_2	Gem (2), Shield (1)
T_3	Elixir (2), Sword (2)
T_4	Elixir (1), Shield (1), Wand (1)
T_5	Elixir (2), Sword (3)
T_6	Shield (1), Wand (1)
T_7	Elixir (1), Shield (3), Sword (1), Wand (2)
T_8	Elixir (5), Giant Wand (3)

Item	Price
Elixir	1
Gem	5
Giant Wand	5
Shield	1
Sword	1
Wand	4

- **Quantity** $q(a, T_i)$ and **price** $p(a)$ of item a
- **Utility of an item** in a transaction: $u(a, T_i) = q(a, T_i) \cdot p(a)$
  - $u(\text{Wand}, T_7) = 2 \cdot 4 = 8$
- **Utility of an itemset** in a transaction: $u(X, T_i) = \sum_{a \in X} u(a, T_i)$
  - $u(\{\text{Elixir}, \text{Wand}\}, T_7) = 1 \cdot 1 + 2 \cdot 4 = 9$

Utility (2): Itemset in the Database

	Items (quantity)	$u(X, T_i)$
T_1	Elixir (1), Shield (1)	
T_2	Gem (2), Shield (1)	
T_3	Elixir (2), Sword (2)	
T_4	Elixir (1), Shield (1), Wand (1)	$1 \cdot 1 + 1 \cdot 4 = 5$
T_5	Elixir (2), Sword (3)	
T_6	Shield (1), Wand (1)	
T_7	Elixir (1), Shield (3), Sword (1), Wand (2)	$1 \cdot 1 + 2 \cdot 4 = 9$
T_8	Elixir (5), Giant Wand (3)	
		$u(X) = 14$

Item	Price
Elixir	1
Gem	5
Giant Wand	5
Shield	1
Sword	1
Wand	4

- **Utility of an itemset**: $u(X) = \sum_{T_i \supseteq X} u(X, T_i)$
  - Sum over all transactions that contain X
- Example: $X = \{\text{Elixir}, \text{Wand}\}$, contained in T_4 and T_7

$$u(X) = u(X, T_4) + u(X, T_7) = 5 + 9 = 14$$

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Utility (3): Transaction Utility

	Items (quantity)	$\text{tu}(T_i)$
T_1	Elixir (1), Shield (1)	2
T_2	Gem (2), Shield (1)	11
T_3	Elixir (2), Sword (2)	4
T_4	Elixir (1), Shield (1), Wand (1)	6
T_5	Elixir (2), Sword (3)	5
T_6	Shield (1), Wand (1)	5
T_7	Elixir (1), Shield (3), Sword (1), Wand (2)	13
T_8	Elixir (5), Giant Wand (3)	20
	Total utility	**66**

Item	**Price**
Elixir	1
Gem	5
Giant Wand	5
Shield	1
Sword	1
Wand	4

- **Transaction utility**: $\text{tu}(T_i) = \sum_{a \in T_i} u(a, T_i)$
  - Total revenue of the whole transaction
  - $\text{tu}(T_7) = 1 \cdot 1 + 3 \cdot 1 + 1 \cdot 1 + 2 \cdot 4 = 13$
- **Goal**: find all **high utility itemsets** (HUI) with $u(X) \geq U_{min}$

Why Apriori Does Not Work Here

- Example: $U_{min} = 14$
  - $u(\{\text{Elixir}\}) = 1+2+1+2+1+5 = 12 < U_{min}$
  - $u(\{\text{Elixir}, \text{Wand}\}) = 14 \geq U_{min}$

$\Rightarrow$ A subset of a high utility itemset is **not** necessarily high utility

- **Support** can only shrink when items are added
- **Utility** can grow: the new item adds its own value

$\Rightarrow$ Pruning with $u(X) < U_{min}$ would lose results

- Idea: prune with an **upper bound** that can only shrink

TWU (1): Upper Bound

- **Transaction-weighted utility**: sum of the *whole* transaction utilities of all transactions containing X

$$\mathrm{TWU}(X) = \sum_{T_i \supseteq X} \mathrm{tu}(T_i)$$

- **Upper bound**: $u(X) \leq \mathrm{TWU}(X)$
  - X is only a **part** of T_i: $u(X, T_i) \leq \mathrm{tu}(T_i)$
  - Sum over the same transactions $\Rightarrow u(X) \leq \mathrm{TWU}(X)$

$X = \{\text{Elixir}, \text{Wand}\}$	T_4	T_7	**Sum**
$u(X, T_i)$: part of the basket	5	9	$u(X) = \mathbf{14}$
$\mathrm{tu}(T_i)$: whole basket	6	13	$\mathrm{TWU}(X) = \mathbf{19}$

TWU (2): Downward Closure

- **Downward closure**: $X \subseteq Y \Rightarrow \mathrm{TWU}(Y) \leq \mathrm{TWU}(X)$
  - Every transaction containing Y also contains X
  - TWU sums whole baskets: fewer baskets, smaller sum
- Example
  - $\mathrm{TWU}(\{\text{Elixir}\}) = 2+4+6+5+13+20 = 50$ (6 baskets)
  - $\mathrm{TWU}(\{\text{Elixir}, \text{Wand}\}) = 6+13 = 19$ (2 baskets)

Consequence

If $\mathrm{TWU}(X) < U_{min}$, then no superset $Y \supseteq X$ can be a high utility itemset:

$$u(Y) \underbrace{\leq}_{\text{upper bound}} \mathrm{TWU}(Y) \underbrace{\leq}_{\text{downward closure}} \mathrm{TWU}(X) < U_{min}$$

Two-Phase Algorithm

- **Phase 1**: Apriori with $\mathrm{TWU}(X) \geq U_{min}$ instead of $\mathrm{supp}(X) \geq S_{min}$
  - Finds all candidates, safe pruning thanks to downward closure
- **Phase 2**: one database scan computes the exact utility $u(X)$ of each candidate

Procedure `TwoPhase`(D, U_{min})**:**
    phase 1: $\hat{H} \leftarrow$ `Apriori`(D, U_{min}) with TWU for pruning
    phase 2: $H \leftarrow \{X \in \hat{H} \mid u(X) \geq U_{min}\}$
    return H

- Liu, Liao, Choudhary: *A Fast High Utility Itemsets Mining Algorithm*, 2005

Walk-through (1): Phase 1, Prune with TWU

Keep X as candidate if $\mathrm{TWU}(X) \geq U_{min} = 14$

$k = 1$	TWU
{Elixir}	50
{Shield}	37
{Wand}	24
{Sword}	22
{Giant Wand}	20
{Gem}	11

$k = 3$	TWU
{Elixir, Shield, Wand}	19

TWU $\geq$ 14: candidate
TWU $<$ 14: rejected

$k = 2$	TWU
{Elixir, Shield}	21
{Elixir, Sword}	22
{Elixir, Wand}	19
{Elixir, Giant Wand}	20
{Shield, Wand}	24
{Shield, Sword}	13
{Sword, Wand}	13
{Shield, Giant Wand}, ...	0

- $k = 3$: {Elixir, Shield, Sword} is pruned, as its subset {Shield, Sword} was rejected
- Only {Elixir, Shield, Wand} remains $\rightarrow$ 11 candidates for phase 2

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Walk-through (2): Phase 2, Filter with u

Keep candidate X if $u(X) \geq U_{min} = 14$

Candidate	TWU	u
{Elixir}	50	12
{Shield}	37	7
{Wand}	24	16
{Sword}	22	6
{Giant Wand}	20	15
{Elixir, Shield}	21	8
{Elixir, Sword}	22	11
{Elixir, Wand}	19	14
{Elixir, Giant Wand}	20	20
{Shield, Wand}	24	21
{Elixir, Shield, Wand}	19	18

$u \geq 14$: high utility $u < 14$: rejected

- 11 candidates → 6 high utility itemsets
- {Elixir}: TWU = 50 passes phase 1, $u = 12$ fails phase 2 (cheap item)
- {Giant Wand}: rare, but valuable
- In frequent itemset mining, {Giant Wand} is not even frequent

High Utility Itemset Mining: Takeaways

	Frequent itemsets	**High utility itemsets**
Data	items	items, quantities, prices
Criterion	$\text{supp}(X) \geq S_{min}$	$u(X) \geq U_{min}$
Downward closure	yes	no $\rightarrow$ use $\text{TWU}(X)$
Algorithm	Apriori	Two-Phase: Apriori on TWU + 1 scan
Top in the game shop	{Elixir}: 0.75	{Shield, Wand}: 21

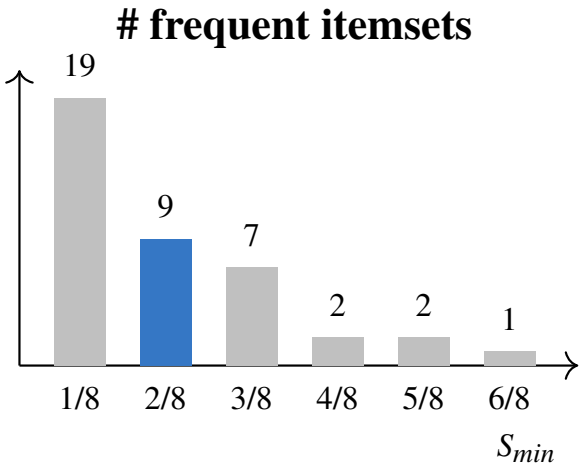
- **Frequent $\neq$ valuable**: {Giant Wand} is rare, but high utility
- **Limitation**: TWU is a loose bound $\rightarrow$ many candidates
- **Newer algorithms** with tighter bounds, no candidates
  - E.g. HUI-Miner (Liu, Qu 2012), EFIM (Zida et al. 2015)

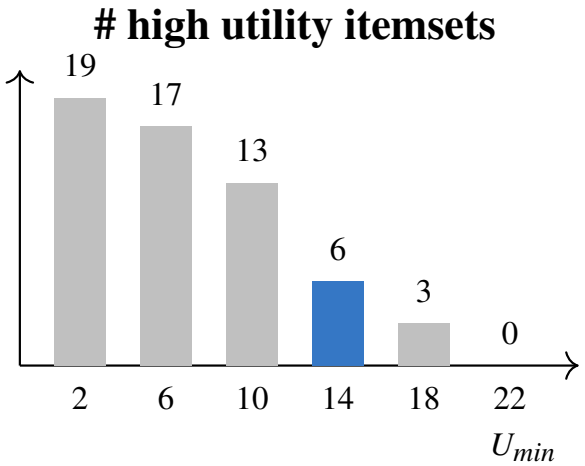
Frequency or Utility?

Question	**Method**	**Game shop**
What is bought **often**?	Frequent itemsets	{Elixir}: in 75 % of baskets
What brings **revenue**?	High utility itemsets	{Shield, Wand}: 21 of 66

- No method is right in general: it depends on the **business question**
- Both results are useful
  - Elixir: draws players into the shop, keep it visible
  - Shield and Wand: bundle with the highest revenue
- **Utility is a design choice**: revenue, profit, margin, ...
- Often combined: high utility **and** enough support
  - {Elixir, Giant Wand}: $u = 20$, but only 1 basket

Choosing Thresholds (1): Effect





■ threshold used in this lecture

- **Too high**: few, mostly obvious patterns
- **Too low**: flood of patterns, runtime grows (up to $2^{|A|}$ itemsets)
- Even here: $S_{min} = 1/8$ instead of $2/8$ more than doubles the result ($9 \rightarrow 19$)

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Choosing Thresholds (2): Strategies

- **No universally correct value**: depends on data and goal
- **Domain knowledge**
  - E.g. minimum revenue that justifies a campaign
  - E.g. minimum number of baskets as evidence
- **Relative thresholds**
  - E.g. U_{min} as share of the total utility: $14/66 \approx 21\,\%$
- **Iterate**: start high, lower step by step until the result is manageable
- **Top-k mining**: ask for the k best itemsets instead of a threshold
- **Check stability**: small changes of the threshold should not change everything

Summary

- **Association rule** $X \to Y$: confidence = empirical $p(Y \mid X)$
- **Rule extraction**: split each frequent itemset, no database scan
- **Lift**: compares confidence with $\text{supp}(Y)$, filters misleading rules
- **Support–confidence plot**: strong rules and niche rules
- **High utility itemsets**: value instead of frequency
- **TWU**: upper bound with downward closure, makes Apriori usable again

Next Steps

- **Assignment 1**: deadline 17 Nov 2026, 23:59 (CET)
- **Next lecture** (11 Nov): Latent Pattern Mining I, recommender systems and matrix factorization